

Demo-performance

# Portfolio performance—S5

- Use the portfolio sorted by momentum:
  - Mom\_perf.csv created in S4
- Calculate the following performance statistics:
  - Quarterly returns of top (TOP) and bottom (BOT) decile portfolios, their return difference (DIF) and t-statistic
  - Alphas of TOP, BOT, and DIF, and t-statistic
  - Annualized information ratios (IR) of TOP, BOT, and DIF, based on CAPM

# Performance Statistics

- Raw return
- CAPM Alpha: The return of the portfolio not associated with its risk vis-à-vis the underlying market portfolio. (a.k.a. Jensen's Alpha---story: Black & Jensen & SEC)

$$r_p = \alpha + \beta r_M + \varepsilon$$

- Fama French 3 factor Alpha: The return of the portfolio not associated with its risk vis-à-vis the underlying multi-factors.

$$r_p = \alpha + \beta_1 r_M + \beta_2 r_{SMB} + \beta_3 r_{HML} + \varepsilon$$

# Performance Statistics

- Information Ratio

- Where the denominator is the residual risk ( $\text{Std}(\varepsilon)$ ) or excess risk caused by deviating from the benchmark.

$$IR = \frac{\alpha^B}{\omega}.$$

- Total risk

- standard deviation of raw returns

## Hw5 :

- Compute the following performance statistics for portfolios sorted by size and momentum (perf\_sizemom.csv created in S4)
  - Quarterly raw returns of top (TOP) and bottom (BOT) decile portfolios, their return difference (DIF) and t-statistic
  - FF4 alphas of TOP, BOT, and DIF, and t-statistic
  - Annualized information ratios (IR) of TOP, BOT, and DIF, based on CAPM